

SIH26249 — 02 Features Spec

SIH 2026 — SIH26249

Air Power: Predictive Maintenance & Fleet Availability

Ministry of Defence — Defence Services Staff College

02 — Features Spec (acceptance criteria + priority)

Priority key: **P0** = 36 h must-have (demo breaks without it). **P1** = stretch / demo-if-time. **P2** = deck-only future work (do not build).

P0 — MUST-HAVE

F1. Tomorrow's MC-Rate Forecaster

- **What:** Fleet mission-capable rate forecast for T+24 h ... T+7 d, per-day, with unfilled high-priority sortie count as the headline capability-gap number.
- **Acceptance:** (a) forecast curve renders for 7 days on the dashboard; (b) every day's number is traceable to per-tail states (MC/PMC/NMC) computable on demand; (c) recomputes from current state in one click; (d) unit test: a tail with critical RUL < 24 h forces NMC on day 1.
- **Risk:** MEDIUM — fusing 5 streams into one calibrated number is the hardest modeling task. Mitigation: rule-based aggregator ships first; learned calibration is stretch only.

F2. What-If Maintenance Levers

- **What:** Apply levers to a cloned fleet snapshot — expedite spares, reassign crew hours, authorize cannibalization — and see before/after MC rate, tails changed, unfilled sorties before/after.
- **Acceptance:** (a) each lever is a pure state patch on a cloned snapshot (base state untouched); (b) recompute < 200 ms; (c) every run stored as a `scenario` row and listed; (d) a scenario can be "applied" to become the working plan; (e) UI shows before/after delta side by side.
- **Risk:** MEDIUM — needs a clean scenario-recompute path. Mitigation: build the aggregator as a pure function of state from day one.

F3. NMC-Driver Ranking

- **What:** Binding constraint per NMC tail (NMCM / NMCS / NMCB), ranked by fleet impact.
- **Acceptance:** (a) every NMC tail has exactly one named driver; (b) fleet score = tails grounded by driver × avg mission priority × days grounded in 7-day forecast; (c) top-5 shown as "fix X → recovers Y tails"; (d) unit test on a hand-built state with known answer.
- **Risk:** LOW — aggregations/Pareto.

F4. Cannibalization Trade-Off Advisor

- **What:** For a chosen NMC recipient tail, proposes a donor: donor must be **lower mission priority AND already NMC**; computes net MC effect and adds the action to the debt ledger.
- **Acceptance:** (a) unit test refuses equal/higher-priority donor; (b) unit test refuses an MC donor; (c) advisor shows net effect (recipient gained – donor opportunity cost); (d) every approved action creates a debt-ledger entry with a repayment queue.

- **Risk:** MEDIUM — greedy net-MC heuristic, explicitly NOT full MILP. OR-Tools MILP is stretch only.
- **Framing rule:** pitch as "authorized cannibalization with guardrails and debt repayment" (per GAO, cannibalization is a *symptom of logistics shortfalls* — never endorse the practice as good).

F5. Sortie-Demand Fusion & Mission-Priority Scoring

- **What:** Sortie schedule (mission type, priority, required status) fused with availability; mission-priority scoring weights every decision by priority 1–3.
- **Acceptance:** (a) unfilled high-priority sorties computed per day; (b) driver ranking and cannibalization both consume priority weights; (c) rule-based, documented in config.
- **Risk:** LOW.

F6. Spares Shortfall / MICAP-Risk Predictor

- **What:** For each part: given current RUL distribution and failure horizon vs lead time and on-hand qty, flag MICAP risk (tail grounded awaiting part).
- **Acceptance:** (a) rule: component RUL < lead_time_h AND qty_on_hand == 0 → MICAP risk on that tail; (b) ranked list of at-risk parts with days-to-stockout; (c) "expedite" action maps to the what-if lever.
- **Risk:** MEDIUM.

F7. Per-Prediction Explainability Panel

- **What:** For any tail/component: top contributing features (sklearn feature contributions) + template-generated plain-English reason ("Engine EGT slope in last 30 cycles contributes 41% of risk; RUL 38 h [31–45 h]").
- **Acceptance:** (a) panel renders on tail drill-down; (b) contributions sum to a stated total; (c) template text never invents numbers not in the prediction row.
- **Risk:** LOW.

F8. Immutable Audit & Provenance Log

- **What:** Hash-chained SQLite `audit_log` — every lever applied, scenario created, cannibalization approved is appended with actor, timestamp, entity, detail JSON, and a hash of the previous row.
- **Acceptance:** (a) verify-chain function passes; (b) tamper test (edit one row → chain breaks); (c) UI audit-log tab lists actions in reverse-chron order.
- **Risk:** LOW.

F9. Crew-Capacity Constraint Scheduler

- **What:** 1-day greedy schedule: assign required fix crew-hours to available trade/shift capacity; work that doesn't fit rolls forward and keeps its tail NMC.
- **Acceptance:** (a) schedule never assigns more hours than a trade has available; (b) greedy order = mission priority × driver fleet score; (c) unassigned work listed as backlog with reason.
- **Risk:** MEDIUM — scoped to 1-day greedy only.

P1 — STRETCH (demo only if time)

F10. Per-Mission MC Forecast (FMC/PMC by mission profile) [LOW]

Acceptance: forecast split by mission profile; sums reconcile to the fleet number.

F11. Cannibalization Debt Ledger [LOW]

Acceptance: every cannibalization creates a repayable debt entry; repayment queue visible; settling a debt (part arrives → reinstall) clears it. (Note: P0 F4 creates the entries; F11 is the full ledger UI.)

F12. Repeat/Recur Discrepancy Watchlist [LOW]

Acceptance: components with $\geq N$ repeat discrepancies in a window are flagged; watchlist tab.

F13. Break-Rate→Parts-Demand Predictor [MEDIUM]

Acceptance: from discrepancy history, forecast parts demand per part_no for the next 7 days; compared against on-hand + due-in.

F14. Offline-First Base Deployment Story [MEDIUM]

Acceptance: demo the dashboard read-only with the API server stopped (pre-baked state + cached page); stated openly as read-only replay.

F15. Plain-English Daily Readiness Brief [LOW]

Acceptance: template-generated brief: MC rate today/forecast, top-3 drivers, top-3 recommended actions, unfilled priority sorties.

F16. Deferred-Discrepancy Portfolio Optimizer [MEDIUM]

Acceptance: given limited crew-hours, choose which deferred discrepancies to work to maximize recovered mission-capable hours.

P2 — DECK-ONLY FUTURE WORK (do not build in 36 h)

F17. Back-Shop & Depot Queue Simulator [HIGH risk]

Why not now: needs queuing simulation + repair-time distributions we cannot validate in 36 h.

F18. Federated Multi-Base Learning [HIGH risk]

Why not now: multi-tenant data, privacy, and sync are a project of their own; architecture notes the SQLite→Postgres swap path.

EXPLICIT NON-GOALS (do not build, do not imply)

1. Any sensor hardware or competing HUMS box.
2. Full MILP/ILP fleet optimizer (greedy heuristic instead; OR-Tools MILP is stretch).
3. LSTM/Transformer RUL models.
4. Real IAF data or connectors to real systems (roadmap only).
5. Multi-base federation, RBAC/auth, real-time streaming physics twins.
6. Survival-analysis RUL (modeling upgrade, future work).
7. Any claim the synthetic results transfer to real aircraft (say the opposite, loudly).